Fiscal Year:	FY 2006	**Task Last Updated:**	FY 04/26/2007
PI Name:	Holstein, Gay R Ph.D.		
Project Title:	Anatomical Studies of Central Vestibular Adaptation-Neurolab Completion Experiment		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Developmental biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Neurobiology		
Space Biology Special Category:	None		
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Zip Code:	10029	**Congressional District:**	15
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2000 Space Biology 00-HEDS-01: UNSOL
Start Date:	04/01/2001	**End Date:**	09/30/2006
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA ARC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	STS-107		
Key Personnel Changes/Previous PI:	-		
COI Name (Institution):	Martinelli, Giorgio (Department of Neurology; Mount Sinai School of Medicine) Friedrich, Victor (Fishberg Center for Neurobiology; Mount Sinai School of Medicine)		
Grant/Contract No.:	NCC2-1173		
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Task Description:	The goal of this project is to identify the morphologic alterations in rat cerebellar cortex that result from transient exposure to an altered gravitational environment. The cerebellum is a part of the brain that is critical for coordination, and for maintaining balance and equilibrium. Two specific aims are being pursued. The first aim is to compare the nature and extent of morphologic changes present in the cerebellar cortex of perfusion-fixed flight and ground control rats. The second aim is to quantify parameters of cerebellar morphology in rats following brief exposure to hypergravity. The objective of the proposed research is to identify the cellular bases for the ultrastructural alterations in Purkinje cells that result from brief exposure to an altered gravitational environment. Data from this study are critical for identifying the health- and performance-related risks associated with human spaceflight, and will establish a ground-based model system for evaluating the neurologic impact of repeated exposures to launch and landing stimulus profiles. The results of these studies will provide new information concerning the gravitational conditions that trigger neuronal structural re-organization in the cerebellum.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Data from this study are critical for identifying the health- and performance-related risks associated with human spaceflight, and will establish a ground-based model system for evaluating the neurologic impact of repeated exposures to launch and landing stimulus profiles.
Task Progress:	Our original flight experiment on STS-90 (April 17-May 3, 1998) yielded qualitative anatomical data indicating that there are structural alterations in the cerebellar nodulus of flight rats 24 hr post-launch, including the presence of lamellar bodies in nodular Purkinje cells. However, qualitative observations of the nodulus from flight rats sacrificed on orbit 14 days after launch did not demonstrate these alterations. Our interpretation of these findings was that ultrastructural reorganization occurs in the cerebellar nodulus during short-term exposure to an altered gravitational environment, but that the architecture of the cerebellar molecular layer re-establishes structural stability following sustained exposure to a constant gravitational force. This interpretation was bolstered by our observation that the nodulus of flight rats sacrificed 24 hr after the 18-day shuttle mission appeared similar to that of the samples obtained on Flight Day 2. To further evaluate this hypothesis, rats were placed on a 1 m centrifuge, which was rotated at 50 rpm (300 deg/s) to deliver 2G of centripetal acceleration to the rats at the end of the arm. The rats were held in Plexiglas boxes and were centrifuged continuously for 4 hours. At the conclusion of the centrifugation, the rats were removed from their holders, and the experiment proceeded as above. As hypothesized, structural alterations including lamellar bodies were observed in the Purkinje cells of these rats. Since the tissue harvesting conditions for that experiment were not optimal for the experimental design, it was also conceivable that the findings reflected a tissue processing artifact. To address this possibility, hypotheses concerning the structural lability, stability and integrity of cerebellar Purkinje cells were to be evaluated in a second flight experiment. This experiment was designed to rigorously test the qualitative findings described above, using tissue from perfusion-fixed animals. A single time point, 24 hr post-flight, was to be evaluated using cerebella from animals flown on the STS-107 mission. Unfortunately, that mission ended tragically. After the STS-107 mission, we began to develop protocols for visualization of several cytochemical and molecular markers for Purkinje cells, in anticipation of a possible re-flight of our experiment. These markers included (1) Concanavalin A, which binds to components of the smooth ER of Purkinje cells, including hypolemmal cisternae and lamellar bodies; (2) antibodies against IP3R, which controls the rapid release of Ca2+ from intracellular membrane-bound stores; (3) antibodies against P-type voltage-gated Ca2+ channels, to test the hypothesis that lamellar body-bound P-type voltage-gated Ca2+ channels are upregulated in rat cerebellum 24 hr post-landing; and (4) antibodies against Calbindin D-28k and cytochrome c, to analyze the role of apoptotic mechanisms in the Purkinje cell dendritic alterations. These studies were aimed at identifying the cellular and molecular mechanisms underlying ultrastructural alterations in Purkinje cells following short-term changes in the gravitational environment. We completed numerous pilot studies that established the optimal conditions for successful labeling of the flight tissue with each of these markers. Shortly thereafter, our experiment was removed from the flight list. In an effort to salvage data from the original flight, we developed a quantitative approach for the analysis of cerebellar ultrastructure that is being used retrospectively to study tissue from the Neurolab mission. In general, this approach allows blind point counting of the relevant ultrastructural features such as degenerated profiles, retracted synapses, and lamellar bodies. We are currently using this method to study matched tissues from flight and cage-control rats from the Neurolab mission. Tissue is being assessed from (a) two parasagittal zones of the nodulus that are known to receive direct and indirect projections from the otolith organs, and (b) two crura of the semilunar lobule (lobule 8) of the cerebellum, which serves as an internal control tissue. This lobule is not thought to receive direct (or indirect) vestibular, auditory, or somesthetic information, and thus is less likely to be affected by launch/landing and microgravity conditions than is the nodulus. Since the molecular layer of cerebellar cortex is not homogeneous, three sets of 10 electron micrographs have been taken through the thickness of the molecular layer (from the pia mater to the Purkinje cells) from each region (two areas of the nodulus and two lobules of Lobule 8) from each animal (flight and cage-control). These micrographs are coded so the readers are blind to the region/animal/flight day source of the images. This approach was used initially to compare the cerebella of flight and control animals harvested on Flight Day 2. Once decoded, the results have been found to support our original qualitative observations. Based on this, we have extended the study to include quantitative analyses of the cerebellar cortex from the other three time points harvested in the Neurolab experiment. In addition, we have expanded the number of parameters quantified at all time points, in an attempt to gain further measures of neural reorganization. Since the data acquisition is not yet complete for all time points, the data have not been decoded and therefore the results are not yet known.
Bibliography Type:	Description: (Last Updated: 04/26/2007)
Abstracts for Journals and Proceedings	Holstein GR. "Rats in Space: Ultrastructural Studies of Cerebellar Cortex. " Association for Research in Otolaryngology, Midwinter Research Meeting, 2004. Association for Research in Otolaryngology, Midwinter Research Meeting, 2004. , Mar-2004

Articles in Peer-reviewed Journals	Cohen B, Yakushin SB, Holstein GR, Dai M, Tomko DL, Badakva AM, Kozlovskaya IB. "Vestibular experiments in space." Adv Space Biol Med. 2005;10:105-64. Review. PMID: 16101106 , Jul-2005